

Siddharth Patel

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Research Interests: Computer architecture & AI hardware accelerators · Digital VLSI & SoC design · Memory systems · Efficient & adaptive AI inference and training

Education

Shiv Nadar Institution of Eminence

B.Tech, Electrical & Computer Engineering

Greater Noida, India

Aug 2023 – Expected May 2027

- **CGPA: 8.73/10.0 · Rank: Top 10% of cohort**
- **Student of the Year, Class of 2027 – university-wide recipient**
- **Dean's List – Fall 2023, Spring 2024, Fall 2025**
- *Specialization: Nanoelectronics & Embedded Systems*
- *Undergraduate Thesis: “An Open-Source GPU Microarchitecture: A Patent-Unencumbered Design with Synthesizable RTL” – in progress (2026–2027)*
- *Selected coursework: ASIC: RTL-to-GDS, VLSI Design, Digital Design with FPGAs, Computer Organization & Design, Semiconductor Devices, Deep Learning, Large Language Models, Embedded Systems*

Publications

*(author in **bold**; † corresponding author)*

Journal Articles

1. **S. Patel**† and R. Singh, “AI Hardware Accelerators for Large Language Models: Architectures and the Memory Wall,” *ACM Computing Surveys*, 2026. [**Under review**]
2. **S. Patel**†, A. Awasthi, A. Chaurasia, and R. Singh, “Bitwise-Reproducible Reduction Across Cluster Reshape: A Compiler/Fabric ABI,” *IEEE Transactions on Parallel and Distributed Systems*, 2026. [**Under review**]
3. **S. Patel** and R. Singh, “RMAI: Runtime Multi-Objective Adaptive Inference through Joint Dynamic Sparsity and Precision Scaling on Resource-Constrained IoT Edge Devices,” *IEEE Internet of Things Journal*, 2026. [**Under review**]

Conference Papers

4. **S. Patel**† and R. Singh, “Hardware-Enabled Root of Trust for IoT Security,” in *Proc. IEEE Intl. Conf. Intelligent Systems and Embedded Design (ISED)*, 2025. [**Published**] DOI: [10.1109/ISED67359.2025.11405039](https://doi.org/10.1109/ISED67359.2025.11405039)
5. S. V. Rajeshwaran, O. Singh, **S. Patel**, S. Singhal, and R. Singh, “Design and Implementation of a Smart Power Controller for Sensor Nodes,” in *Proc. Intl. Conf. on Intelligent Technologies for a Sustainable & Inclusive Future (ICITSIF)*, 2026. [**In press**]
6. **S. Patel**† and R. Singh, “Architecture-Aware Pruning and Quantisation for CPU-Only IoT Edge Inference,” in *Proc. IEEE World Forum on the Internet of Things (WF-IoT)*, 2026. [**Under review**]

Research Experience

An Open-Source GPU Microarchitecture (Undergraduate Thesis)

2026 – Present

Undergraduate Researcher · Advisor: Dr. Rohit Singh

Greater Noida, India

- Designing a fully open, patent-unencumbered GPU from first principles – microarchitecture, ISA, and synthesizable RTL – to address the absence of an open, industrial-grade GPU (existing designs are locked behind AMD/NVIDIA IP).
- Releasing as open source for research and educational use.

Hardware for AI — Deterministic Large-Model Inference

Dec 2025 – Present

Undergraduate Researcher · Advisors: Dr. Rakesh Palishtty & Dr. Rohit Singh Greater Noida, India

- Investigating sources of non-determinism in large-model inference on parallel hardware, and co-designing compiler- and fabric-level mechanisms to eliminate them.
- Designed a compiler/fabric ABI that **guarantees bitwise-reproducible reductions across heterogeneous GPU cluster reshapes** – bit-identical results independent of how the cluster is partitioned or resized. ($\rightarrow$ *IEEE TPDS [under review]*)
- Prototyping compute-in-memory primitives and alternative number systems for deterministic computation.

Edge-AI Optimization — Adaptive & Architecture-Aware Inference

Oct 2025 – May 2026

Undergraduate Researcher · Advisor: Dr. Rohit Singh

Greater Noida, India

- Designed **RMAI**, a runtime adaptive-inference framework that jointly tunes per-layer weight sparsity and quantization precision to live device state (battery, thermal, memory). ($\rightarrow$ *IEEE IoT Journal [under review]*)
- Demonstrated up to **14.8% lower latency and 15.4% energy savings** on ResNet-50 vs. a static INT8 baseline (Raspberry Pi 4), with **38 $\times$ greater thermal robustness** under sustained load; validated across ResNet-50, MobileNetV2, DeiT-Small, and YOLOv8m.
- Developed **architecture-aware pruning and quantization for CPU-only edge inference**. ($\rightarrow$ *IEEE WF-IoT [under review]*)

Smart Power Controller for Sensor Nodes

Dec 2024 – Oct 2025

Research Assistant · Advisor: Dr. Rohit Singh

Greater Noida, India

- Contributed to the design of an energy-efficient wireless sensor node with a programmable power-control unit for smart-irrigation / smart-city deployments.
- Implemented demand-based power management that **doubled battery life** for power-hungry sensor nodes. ($\rightarrow$ *ICITSIF [in press]*)

Hardware-Enabled Root of Trust for IoT

Aug 2023 – Dec 2024

Undergraduate Researcher · Advisor: Dr. Rohit Singh

Greater Noida, India

- Designed a hardware root-of-trust with immutable device identities and RSA-OAEP secure boot to resist spoofing and tampering.
- Achieved **99.8% authentication accuracy** under simulated attack scenarios. ($\rightarrow$ *IEEE ISED [published]*)

Industry Experience

Technology R&D (TRD) Intern

Jun 2026 – Present

STMicroelectronics

Greater Noida, India

- Building **AI-driven automation for STMicroelectronics' end-to-end memory-design workflow**, replacing a manual, multi-engineer, per-variant RTL process with an intelligent generation pipeline.
- Applying **ML-guided design-space exploration and learned optimization**, and **fine-tuning LLMs** for domain-specific accuracy on RTL tasks.
- Developing **LLM-assisted RTL generation** and architecting a **domain-specific AI agent from scratch** to drive the flow end-to-end.
- Produces synthesis-ready, **ECC-protected memory subsystems** (encoder/decoder RTL; SEC/SECDED) across 7+ configuration axes, byte- to gigabyte-scale.

Independent Consultant — Electronics Design

Feb 2025 – Jul 2025

[Electronics Manufacturer – add name if shareable]

Delhi NCR, India

- **Sole engineer** on a full product redesign – re-architected 3-/4-phase voltage stabilizers end-to-end (circuit, enclosure, PCB, firmware) and delivered independently, ahead of schedule.
- Cut bill-of-materials and assembly complexity via modular reuse (**$\sim$ 70% lower production cost**), with backward compatibility for legacy systems.

Teaching Experience

Curriculum Designer & Lab Instructor — Electronics & IoT Jul 2025 – May 2026
Department of Electrical Engineering, Shiv Nadar Institution of Eminence *Greater Noida, India*

- Designed lab experiments, modules, learning objectives, and assessments for the first-year Electronics & IoT curriculum (**480+ students**).
- Instructed **120+ students** in hands-on electronics / IoT lab sessions.
- Led build-out of a 20-station Electronics Research & Innovation Lab (equipment procurement + setup).

Honors & Awards

Student of the Year, Class of 2027 – sole university-wide recipient across all departments, Shiv Nadar Institution of Eminence (2026)

Selected from 20,000+ applicants nationwide – Anveshana National Science Fair (2025)

Atal Incubation Centre Scholar – competitively selected for the *Life Link* health platform; awarded a grant (INR 100K) with mentorship and incubation support (2024–25)

Invited Peer Reviewer – IEEE Intl. Conf. on Next-Generation Adaptive Research & Innovations (ICNARI 2026)

Mathematics Olympiad – Gold Medal (national level, 2019)

Technical Skills

Languages: Python, C/C++, Verilog, SystemVerilog, VHDL, Amaranth HDL, Tcl

Computer Architecture & Digital Design: RISC-V, CPU Microarchitecture, ASIC/RTL Design, FPGA Design, Parameterized RTL Generation

Memory & VLSI: SRAM/DRAM Architecture, Memory IP Development, Compute-in-Memory (CIM), ECC & Coding Theory (SEC/SECDED, BCH, Hamming), Formal Verification

AI/ML: PyTorch, Custom ML Library Development, Transformer/LLM Architectures, LLM Fine-Tuning, LLM-Assisted RTL Generation, AI Agents, ML-Guided Design-Space Exploration, Mixed-Precision Quantization, Dynamic Sparsity

EDA Tools: Synopsys (RTL-to-GDS flow), Cadence Virtuoso & Xcelium, Xilinx Vivado, KiCad

Embedded: STM32, Arduino, Raspberry Pi, Embedded Linux

Outreach

Anveshana National Science Fair – led STEM workshops for 50+ students and presented a secure-IoT prototype to 300+ attendees, Pragati Maidan, Delhi (2024–25).